

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) / S2 (PT) (R,S) Examination June 2023 (2019 Scheme)

Course Code: EET206**Course Name: DIGITAL ELECTRONICS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

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|----|--|---|
| 1 | Convert 10110010 to hexadecimal and Octal | 3 |
| 2 | Draw the truth table and explain the operation of EX-OR and EX-NOR gates | 3 |
| 3 | Get the dual of the expression $AB + \overline{AC} + A\overline{B}C(AB + C) = 1$ | 3 |
| 4 | Derive the expression for sum and carry of a full adder | 3 |
| 5 | Explain a two bit comparator | 3 |
| 6 | Draw the circuit of a 1 to 4 Demultiplexer and explain | 3 |
| 7 | Explain the truth table and excitation table of a JK flip flop | 3 |
| 8 | What are the asynchronous inputs of a flip flop? Why are they called so? | 3 |
| 9 | Draw the state diagram of a three bit binary down counter | 3 |
| 10 | What is FPGA? | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|--|---|
| 11 | a) Perform the following (i) 165.875_{10} to binary (ii) $4BAC_{16}$ to binary (iii) 378_{10} to octal | 7 |
| | b) What are universal gates. Why are they called so? | 7 |
| 12 | a) With the help of a neat diagram explain CMOS NAND gate | 7 |
| | b) Describe error detection and correction using parity checking | 7 |

Module -2

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|----|--|---|
| 13 | a) Design and set up a half adder/subtractor circuit with mode control. | 7 |
| | b) Obtain the simplified POS expression using K-map – $Y = \sum m(1,3,7,11,15) + d(0,2,4)$ | 7 |
| 14 | a) Using a neat circuit explain a 3 bit carry look ahead adder | 9 |
| | b) Write and explain any five basic laws of Boolean algebra | 5 |

Module -3

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|----|----|--|----|
| 15 | a) | Implement $f(x,y,z) = \sum m(1,2,6,7)$ using 4 x 1 MUX | 7 |
| | b) | Explain a 3 to 8 decoder | 7 |
| 16 | a) | Design and set up a Binary to BCD converter circuit | 10 |
| | b) | How is priority encoder different from encoder | 4 |

Module -4

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|----|----|--|---|
| 17 | a) | Draw the logic circuit of SR flip flop and explain it | 7 |
| | b) | What is modulus of a counter. Draw the circuit of an asynchronous decade counter | 7 |
| 18 | a) | Enumerate the steps for design of a synchronous counter | 7 |
| | b) | Draw the circuit of 4 bit ring counter and give the timing diagram | 7 |

Module -5

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|----|----|---|----|
| 19 | a) | Design a 4 bit weighted resistor DAC whose full scale output voltage is -5V. The logic levels are Logic 1 = +5 V and logic 0 = 0 V. What is the output voltage when the input is 1101 | 10 |
| | b) | Give the Verilog code for AND gate | 4 |
| 20 | a) | Explain Flash type A/D converter | 10 |
| | b) | Differentiate PAL and PLA | 4 |
